Orthogonal Phase Modulation LFM Waveforms for MIMO Distributed Space-Based Radar

Yuxing Zhao
School of Integrated Circuits
Shanghai Jiao Tong University
Shanghai, China
zyx_0303@sjtu.edu.cn

Changrong Hu
Shanghai Academy of Spaceflight Technology
Shanghai, China
1074986429@qq.com

Haojuan Yuan
Shanghai Aerospace Electronic Technology Institute
Shanghai, China
yuanhaojuan@126.com

Muyang Zhan
Shanghai Aerospace Electronic Technoology Institute
Shanghai, China
zhanmuyang1992@126.com

Qing Ling
College of Electronic Engineering
Naval University of Engineering
Wuhan, China
ling_qing1997@163.com

Fan Yang
School of Integrated Circuits
Shanghai Jiao Tong University
Shanghai, China
yfqq-23@sjtu.edu.cn

Lang Xia
School of Integrated Circuits
Shanghai Jiao Tong University
Shanghai, China
xl2023@sjtu.edu.cn

Zihao Zou
Aerospace Information Research Institute
Chinese Academy of Sciences
Beijing, China
zouzihao@aircas.ac.cn

Penghui Huang*
School of Integrated Circuits
Shanghai Jiao Tong University
Shanghai, China
huangpenghui@sjtu.edu.cn

***Abstract*—Due to constraints on satellite platform size, payload capacity, and power–aperture product, the air moving target indication (AMTI) performance of conventional single-platform spaceborne radars is fundamentally limited, especially for slow and weak targets. Distributed space-based radar (DSBR) systems mitigate this limitation by exploiting extended along-track baselines. However, the target detection performance is restricted by low transmit channel utilization in single-transmit multiple-receive architectures. This paper proposes a MIMO-DSBR architecture based on pulse-to-pulse orthogonal phase sequence modulation, where DFT-based orthogonal sequences are applied to LFM waveforms to enable effective channel separation. Simulation results demonstrate that the proposed scheme significantly improves channel utilization and, under ideal orthogonality conditions, reduces the minimum detectable velocity (MDV) from 9.4 m/s to 4.6 m/s while achieving approximately a 3 dB improvement in output signal-to-clutter-plus-noise ratio (SCNR), without increasing system complexity.**

Keywords*—*Distributed space-based radar (DSBR), MIMO, Orthogonal Phase Sequence, Pulse-interval modulation, AMTI

I. Introduction

Compared with ground-based and airborne radar systems [1], space-based radar offers wide-area coverage, all-weather capability, and long-duration persistent surveillance, making it well suited for AMTI [2]. However, the AMTI performance of conventional single-platform spaceborne radar is fundamentally limited by platform size, payload capacity, and the achievable power–aperture product [2], [3]. In particular, the short along-track baseline (ATB) leads to a high minimum detectable velocity (MDV), severely degrading the detection of slow-moving and weak targets.

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To overcome these limitations, DSBR has attracted increasing attention [3], [4]. By coordinating multiple satellites, DSBR systems synthesize a virtual aperture with an extended ATB, providing additional spatial degrees of freedom and improved clutter suppression. Benefiting from spatial power synthesis and multichannel processing, DSBR can achieve superior AMTI performance and reduced MDV compared with single-platform systems.

Nevertheless, most existing DSBR architectures adopt a single-transmit multiple-receive (STMR) configuration, in which only one satellite actively transmits. Although this simplifies synchronization and waveform design, the transmit channel utilization efficiency is low, limiting the achievable system gain and waveform diversity. To further exploit the potential of distributed platforms, multiple-input multiple-output (MIMO) techniques have been introduced, allowing multiple satellites to transmit simultaneously [5]. However, MIMO-based DSBR systems require highly orthogonal waveforms to avoid inter-channel interference [6], [7], which is challenging due to long inter-satellite baselines and relatively high platform velocity.

Inspired by the work of previous researchers, a MIMO-DSBR architecture based on pulse-to-pulse orthogonal phase sequence modulation is proposed in this paper. Discrete Fourier transform (DFT)-based orthogonal phase sequences are employed to modulate the transmitted waveforms of the master and auxiliary satellites, ensuring waveform orthogonality within the MIMO framework. At the receiver, corresponding phase compensation enables effective demodulation and channel separation. Simulation results demonstrate that the proposed architecture significantly improves transmit channel utilization and achieves notable MDV performance enhancement without increasing system complexity.

The remainder of this paper is organized as follows. Section II presents the DSBR signal model. Section III describes the theoretical framework of the proposed MIMODSBR architecture. Section IV provides simulation results and performance analysis. Finally, Section V concludes the paper.

II. Distributed Signal Model

The three-dimensional geometric model for AMTI in a DSBR system operating in a single-transmit dual-receive configuration is illustrated in Fig. 1. In this system, the master satellite transmits linear frequency modulated (LFM) radar pulses, while both the master and auxiliary satellites receive the target echoes. The antennas of the master and auxiliary satellites are identical, and the azimuth antenna of each satellite is uniformly divided into N equivalent channels. The spacing between adjacent equivalent phase centers of a single satellite is denoted by d, and the platform velocity is V_a.

A right-handed Cartesian coordinate system is established as shown in Fig. 1, where the Y-axis represents the flight direction of the satellite platform, the Z-axis is perpendicular to the ground, and the X-axis, together with the Y- and Z-axes, forms a right-handed coordinate system. A hybrid baseline exists between the master and auxiliary satellites, whose components along the X-, Y-, and Z-axes are denoted by B_x, B_y, and B_z, respectively. Here, B_x represents the along-track baseline component, while B_y and B_z denote the vertical baseline components along the Y- and Z-axes, respectively. The angles $\theta_{EL,t}$ and $\theta_{AZ,t}$ denote the depression angle and azimuth angle of the observed target with respect to the master satellite radar, respectively.

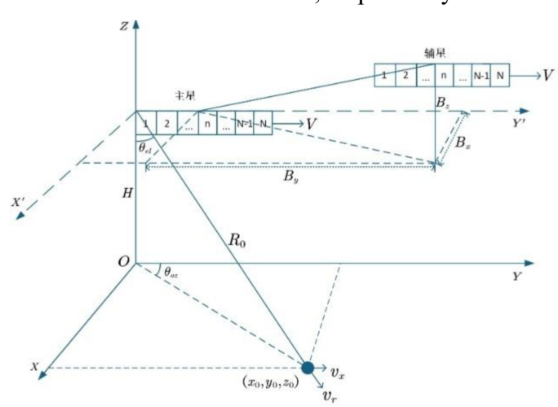

Fig. 1 Three-dimensional observation map of moving targets by distributed satellite-borne radar

The target is located at the coordinate $(x_0+v_x t_m, y_0+v_y t_m, z_0)$, and the radial velocity of the moving target is denoted by $v_r=\frac{x_0}{R_0}v_x+\frac{y_0}{R_0}v_y$. The coordinates of the nth equivalent channel of the master and auxiliary satellites are denoted by $\left(0, Vt_m+d_{pc\_n}, H\right)$ and $\left(\frac{B_x}{2}, Vt_m+d_{pc\_n}+\frac{B_y}{2}, H+\frac{B_z}{2}\right)$, respectively. Here, H represents the satellite altitude, $d_{pc\_n}=\frac{(N-3)d}{4}+\frac{n}{2}d$ denotes the along-track spacing of the nth equivalent phase center for both the master and auxiliary satellites.

Assume that the master satellite transmits a LFM signal, whose baseband expression is given by:

$$P(t)=\operatorname{rect}\left(\frac{t}{T_p}\right)\exp\left(j\pi k t^2\right)\exp\left(j2\pi f_0 t\right) \tag{1}$$

Here, $rect(x)=\begin{cases}1, & |x|\le 1/2\\ 0, & otherwise\end{cases}$ denotes the rectangular window function, while t, T_p, k, and f_0 represent the fast-time variable, pulse duration, range chirp rate, and carrier frequency, respectively. Based on the equivalent phase center model, and neglecting the effect of terrain undulation, the equivalent two-way instantaneous slant ranges between the nth receive channel of the master and auxiliary satellites and the moving target located at (x_0, y_0, z_0) can be, respectively, expressed as:

$$R_{t1}\left(t_m\right)\approx\sqrt{\left(x_0+v_x t_m\right)^2+\left(d_{pc\_n}+Vt_m-y_0-v_y t_m\right)^2+H^2} \tag{2}$$

$$R_{t2}\left(t_m\right)\approx\sqrt{\begin{array}{l}\left(\frac{B_x}{2}-x_0-v_x t_m\right)^2+\left(H+\frac{B_z}{2}\right)^2\\+\left(d_{pc\_n}+\frac{B_y}{2}+Vt_m-y_0-v_y t_m\right)^2\end{array}} \tag{3}$$

By performing a Taylor series expansion of the slant range expressions in (2) and (3), and compensating for the second-order phase error using the system parameters, the clutter and target echo signals of the master and auxiliary satellites, after downconversion and range pulse compression, can be, respectively, expressed as (4) and (5).

$$\begin{aligned}s_{t1}\left(t,t_m\right)\approx{}&\sigma_{t1}\operatorname{sinc}\left[B\left(t-\frac{2R_0}{c}\right)\right]\operatorname{rect}\left(\frac{t_m}{T_a}\right)\operatorname{rect}\left(\frac{t_m+(n-1)d/2V}{T_a}\right)\\&\times\exp\left\{-j\frac{4\pi}{\lambda}\left(v_r t_m-Vt_m\cos\theta_{\text{cone,t}}+\frac{d_{pc\_n}\left(V-v_y\right)t_m}{R_0}\right)\right\}\\&\times\exp\left\{-j\frac{4\pi}{\lambda}\left[R_0-d_{pc\_n}\cos\theta_{\text{cone},t}\right]\right\}\end{aligned} \tag{4}$$

$$\begin{aligned}S_{t2}(t,t_m)\approx{}&\sigma_{t2}\operatorname{sinc}\left[B\left(t-\frac{2R_0}{c}\right)\right]\operatorname{rect}\left(\frac{t_m}{T_a}\right)\operatorname{rect}\left(\frac{t_m+(n-1)d+B_y/2V}{T_a}\right)\\&\exp\left\{-j\frac{4\pi}{\lambda}\left[R_0-\left(d_{pc\_n}\cos\theta_{cone,t}+\frac{B_y}{2}\cos\theta_{cone,t}\right)\right]\right\}\\&\exp\left\{-j\frac{4\pi}{\lambda}\left[\begin{array}{l}v_r t_m-Vt_m\cos\theta_{cone,t}+\frac{B_y(V-v_y)t_m}{2R_0}\\+\frac{d_{pc\_n}(V-v_y)t_m}{R_0}-\frac{B_x v_x t_m}{2R_0}\end{array}\right]\right\}\end{aligned} \tag{5}$$

where σ_i, B, c, λ and t_m represent the pulse-compressed echo amplitude, signal bandwidth, speed of light, signal wavelength, and slow-time variable for the primary/secondary stars, respectively.

III. Orthogonal Phase Modulation-Based MIMO-DSBR

Based on the DSBR signal model in (1)-(5), a distributed MIMO architecture is introduced in which both the master and auxiliary satellites simultaneously transmit orthogonally modulated LFM waveforms to improve channel utilization and AMTI performance. By exploiting the geometric equivalence of the cross-transmit–receive paths, the proposed configuration introduces an additional equivalent virtual channel, effectively increasing the system degrees of freedom.

A. Orthogonal Phase Sequence Modulation

The master and auxiliary satellites transmit LFM signals given by:

$$P_i(t,t_m)=\mathrm{rect}\left(\frac{t}{T_p}\right)\exp\left(j\pi kt^2\right)\exp\left(j2\pi f_0 t\right)\times\exp(j\varphi_i t_m) \tag{6}$$

where φ_i denotes the orthogonal phase sequence imposed on the transmitted signals of the master and auxiliary satellites to ensure orthogonality among the MIMO channels.

In this paper, DFT-based orthogonal phase sequences are employed, which can be expressed as:

$$\varphi_i=2\pi k_i n/N_{\mathrm{a}},\mathrm{n}=0,\ldots,\mathrm{N_a}\text{-}1 \tag{7}$$

where N_a denotes the number of azimuth pulses.

The orthogonality of the DFT-based phase sequences can be demonstrated through inner product operations. Specifically, the inner product between two sequences corresponding to $k=k_1$ and $k=k_2$ is calculated as:

$$\begin{aligned}\left\langle s_{k_1},s_{k_2}\right\rangle&=\sum_{n=0}^{N-1}\exp\left(j\frac{2\pi}{N}k_1 n\right)\exp\left(-j\frac{2\pi}{N}k_2 n\right)\\&=\sum_{n=0}^{N-1}\exp\left(j\frac{2\pi}{N}(k_1-k_2)n\right)\end{aligned} \tag{8}$$

Since $\sum_{n=0}^{N-1}\exp\left(j\frac{2\pi}{N}(k_1-k_2)n\right)=\begin{cases}N, & k_1=k_2\\ 0, & k_1\neq k_2\end{cases}$, we select $k_1\neq k_2$,the inner product of the two sequences becomes zero, i.e., $\left\langle s_{k_1},s_{k_2}\right\rangle=0,k_1\neq k_2$, which satisfies the requirement of statistical orthogonality in the mathematical sense.

B. MIMO-DSBR Moving Target Echo Signal Model

Under the MIMO-DSBR architecture, the equivalent slant ranges for the master and auxiliary monostatic paths and the master–auxiliary bistatic path are given in (9), (10), (11), respectively.

$$R_{AAn}\left(t_m\right)\approx\sqrt{\left(x_0+v_x t_m\right)^2+\left(d_{pc\_n}+Vt_m-y_0-v_y t_m\right)^2+H^2} \tag{9}$$

$$R_{BBn}\left(t_m\right)\approx\sqrt{\begin{aligned}&\left(B_x-x_0-v_x t_m\right)^2+\left(H+B_z\right)^2\\&+\left(d_{pc\_n}+B_y+Vt_m-y_0-v_y t_m\right)^2\end{aligned}} \tag{10}$$

$$R_{\mathrm{AB}n}\left(t_m\right)\approx\sqrt{\begin{aligned}&\left(\frac{B_x}{2}-x_0-v_x t_m\right)^2+\left(H+\frac{B_z}{2}\right)^2\\&+\left(d_{pc\_n}+\frac{B_y}{2}+Vt_m-y_0-v_y t_m\right)^2\end{aligned}} \tag{11}$$

where A and B denote the master and auxiliary satellites, respectively. A_n and B_n denote the nth channel of the primary and secondary stars, respectively. The single-satellite radar operates in a full-aperture transmitting and sub-aperture receiving mode.

After downconversion and range pulse compression, the moving target echo signals received by the master and auxiliary satellites can be uniformly expressed as:

$$\begin{aligned}s_{i,n}\left(t,t_m\right)=\sigma_i\,&\mathrm{sinc}\left[B\left(t-\frac{2R_i\left(t_m\right)}{c}\right)\right]\\&\exp\left[-j\frac{4\pi}{\lambda}R_i\left(t_m\right)\right]\exp\left(j\varphi_i t_m\right)\end{aligned} \tag{12}$$

where $i=AA,AB,BB$ correspond to the monostatic master-transmit/master-receive path, the bistatic master-transmit/auxiliary-receive path, and the monostatic auxiliary-transmit/auxiliary-receive path, respectively.

By substituting (9), (10), (11) into (12), the moving target echo signal expression for the MIMO-DSBR system can be obtained. Finally, the echo signals corresponding to the three radar learning paths are, respectively, given by:

$$\begin{aligned}S_{AAn}(t,t_m)=\sigma_{AAn}\,&\mathrm{sinc}\left[B\left(t-\frac{2R_0}{c}\right)\right]\times\mathrm{rect}\left(\frac{t_m}{T_a}\right)\mathrm{rect}\left(\frac{t_m+(n-1)d/2V}{T_a}\right)\\&\times\exp\left\{-j\frac{4\pi}{\lambda}\left(v_r t_m-Vt_m\cos\theta_{\mathrm{cone,t}}+\frac{d_{pc\_n}\left(V-v_y\right)t_m}{R_0}\right)\right\}\\&\times\exp\left\{-j\frac{4\pi}{\lambda}\left[R_0-d_{pc\_n}\cos\theta_{\mathrm{cone},t}\right]\right\}\exp\left(j\varphi_A t_m\right)\end{aligned} \tag{13}$$

$$\begin{aligned}S_{ABn}(t,t_m)\approx\sigma_{ABn}&\mathrm{sinc}\left[B\left(t-\frac{2R_0}{c}\right)\right]\mathrm{rect}\left(\frac{t_m+(n-1)d+B_y/2V}{T_a}\right)\mathrm{rect}\left(\frac{t_m}{T_a}\right)\\&\exp\left\{-j\frac{4\pi}{\lambda}\left\{\begin{aligned}&v_r t_m-Vt_m\cos\theta_{cone,t}+\frac{B_y(V-v_y)t_m}{2R_0}+\\&\frac{d_{pc\_n}(V-v_y)t_m}{R_0}-\frac{B_x v_x t_m}{2R_0}\end{aligned}\right\}\right\}\\&\exp\left\{-j\frac{4\pi}{\lambda}\left[R_0-\left(d_{pc\_n}\cos\theta_{cone,t}+\frac{B_y}{2}\cos\theta_{cone,t}\right)\right]\right\}\exp\left(j\varphi_A t_m\right)\end{aligned} \tag{14}$$

$$\begin{aligned}s_{BBn}(t,t_m)\approx\sigma_{BBn}&\mathrm{sinc}\left[B\left(t-\frac{2R_0}{c}\right)\right]\mathrm{rect}\left(\frac{t_m+(n-1)d}{T_a}\right)\mathrm{rect}\left(\frac{t_m}{T_a}\right)\\&\exp\left\{-j\frac{4\pi}{\lambda}\left[R_0-\left(d_{pc\_n}\cos\theta_{cone,t}+\frac{B_{y1}}{2}\cos\theta_{cone,t}\right)\right]\right\}\\&\exp\left\{-j\frac{4\pi}{\lambda}\left\{\begin{aligned}&v_r t_m-Vt_m\cos\theta_{cone,t}+\frac{B_{y1}(V-v_y)t_m}{2R_0}\\&+\frac{d_{pc\_n}(V-v_y)t_m}{R_0}-\frac{B_{x1}v_x t_m}{2R_0}\end{aligned}\right\}\right\}\exp\left(j\varphi_B\right)\end{aligned} \tag{15}$$

where $B_{x1} = 2B_x$, $B_{y1} = 2B_y$, $B_{z1} = 2B_z$.

IV. Simulation Experiments

In this section, numerical simulations are carried out, and a comparative analysis with the conventional DSBR system is presented. The simulation performance is evaluated in terms of the output signal-to-clutter-plus-noise ratio (SCNR) after clutter suppression and MDV, which are adopted as key metrics to assess the effectiveness of the proposed architecture for AMTI.

TABLE I. Radar System Parameters And Antenna Parameters

Parameters	Value
Number of satellites	2
Number of azimuth channels for each satellite	8
Carrier frequency	1.2GHz
Platform height	506km
Average power	4000W
Signal bandwidth	3MHz
Pulse repetition frequency	4000Hz
Antenna Installation Tilt Angle	45°
Coherent accumulation time	60ms
ATB	100m

The simulation scenario considers a densely formed L-band distributed space-based early warning radar (DSBEWR) system. Some of the key radar system parameters are selected with reference to the U.S. Large L-band Space-Based Radar (LLSBR) system [8], and the detailed parameter configuration is listed in Table I.

A. MIMO-DSBR

Fig. 2 compares the MDV performance of the MIMO-DSBR and conventional DSBR systems under identical system parameters and processing schemes. In the conventional DSBR architecture, sparse inter-satellite spacing induces spatial grating lobes, degrading AMTI performance.

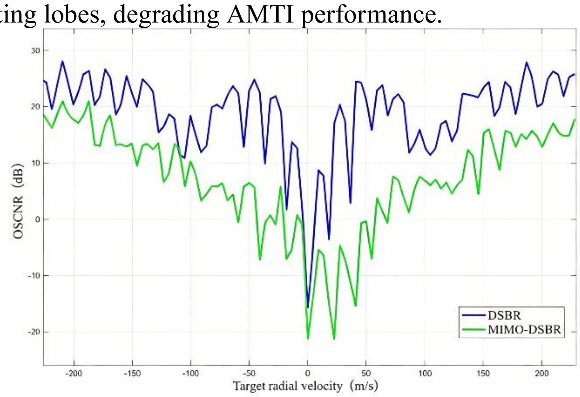

Fig. 2 MIMO System MDV Diagram

Although simultaneous transmission is enabled in the MIMO-DSBR configuration, the limited orthogonality of pulse-to-pulse phase-modulated waveforms leads to insufficient inter-channel isolation, causing clutter leakage and preventing effective MDV improvement.

B. Analysis of the Ideal Case with Fully Orthogonal MIMO Channels

To assess the potential of the MIMO architecture, echo signals are analyzed under ideal conditions assuming perfect orthogonality among MIMO transmit channels, enabling ideal channel separation at the receiver. As shown in Fig. 3, under this assumption the MIMO-DSBR system reduces the minimum detectable velocity from 9.4 m/s to 4.6 m/s compared with the conventional DSBR configuration, while achieving approximately a 3 dB improvement in output SCNR for slow-moving targets. These results indicate that, when waveform orthogonality is guaranteed, the MIMO architecture can substantially enhance low-velocity target detection performance and unlock the full potential of DSBR systems.

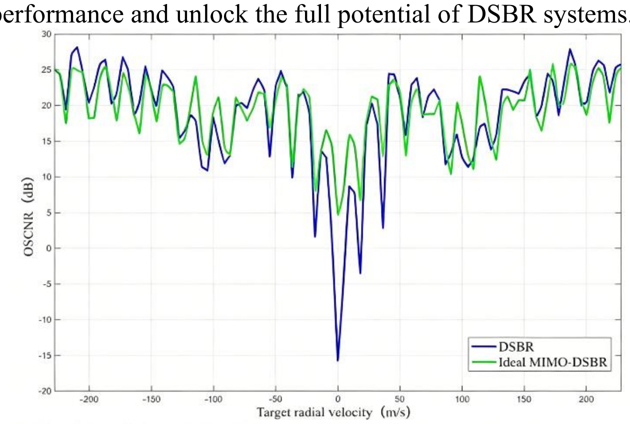

Fig. 3 Ideal Conditions MIMO System MDV Diagram

V. Conclusion

This paper investigated a MIMO-DSBR architecture for improving AMTI performance in DSBR systems. Simulation results show that conventional DSBR suffers from MDV degradation due to spatial grating lobes, while practical MIMODSBR performance is limited by imperfect waveform orthogonality and inter-channel leakage. Under ideal conditions with fully orthogonal MIMO channels, the proposed architecture significantly reduces MDV and improves SCNR for slow-moving targets. These results demonstrate the performance potential of MIMO-DSBR and highlight the importance of waveform orthogonality and grating-lobe suppression in future system design.

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